

* Home Assignment - 5 *

Q37] Following is prices,
 3.88, 3.90, 3.93, 3.90, 3.93, 3.96, 3.88, 3.94, 3.96,
 3.88, 3.94, 3.99, 3.98.

Find range, histogram & box plot. State outliers.
 Find mean, mode Karl Pearson's coeff of skewness.
 Bowley's coeff of skewness, coeff of kurtosis. Hence
 state distribution is tvely / -vely skewed or is
 symmetric. Also state it is platykurtic / meso / kurtic /
 leptokurtic.

→

$$\begin{aligned} \text{Range} &\Rightarrow \text{Max.} - \text{Min} \\ &= 3.99 - 3.88 \\ &= 0.11 \end{aligned}$$

Freq. table $\Rightarrow$

Price (in dollars)	Freq.
3.88	3
3.90	2
3.93	2
3.94	2
3.96	2
3.98	1
3.99	1

$$\begin{aligned} \text{Mean} &= \frac{3.88 + 3.90 + 3.93 + 3.90 + 3.93 + 3.96 + 3.88 + 3.94 + 3.94 + 3.96 + 3.88 + 3.94 + 3.99 + 3.98}{13} \\ &= \frac{51.34}{13} = 3.95 \end{aligned}$$

$$\therefore \text{Mode} = 3.88$$

Karl Pearson's (coeff. of skewness $\Rightarrow$)

$$\text{skewness} = \frac{3(\text{Mean} - \text{Median})}{\text{S.D.}}$$

$$\therefore \text{var} = \frac{\sum (x_i - \text{Mean})^2}{n}$$

$$\therefore \text{standard Deviation} \Rightarrow 0.034$$

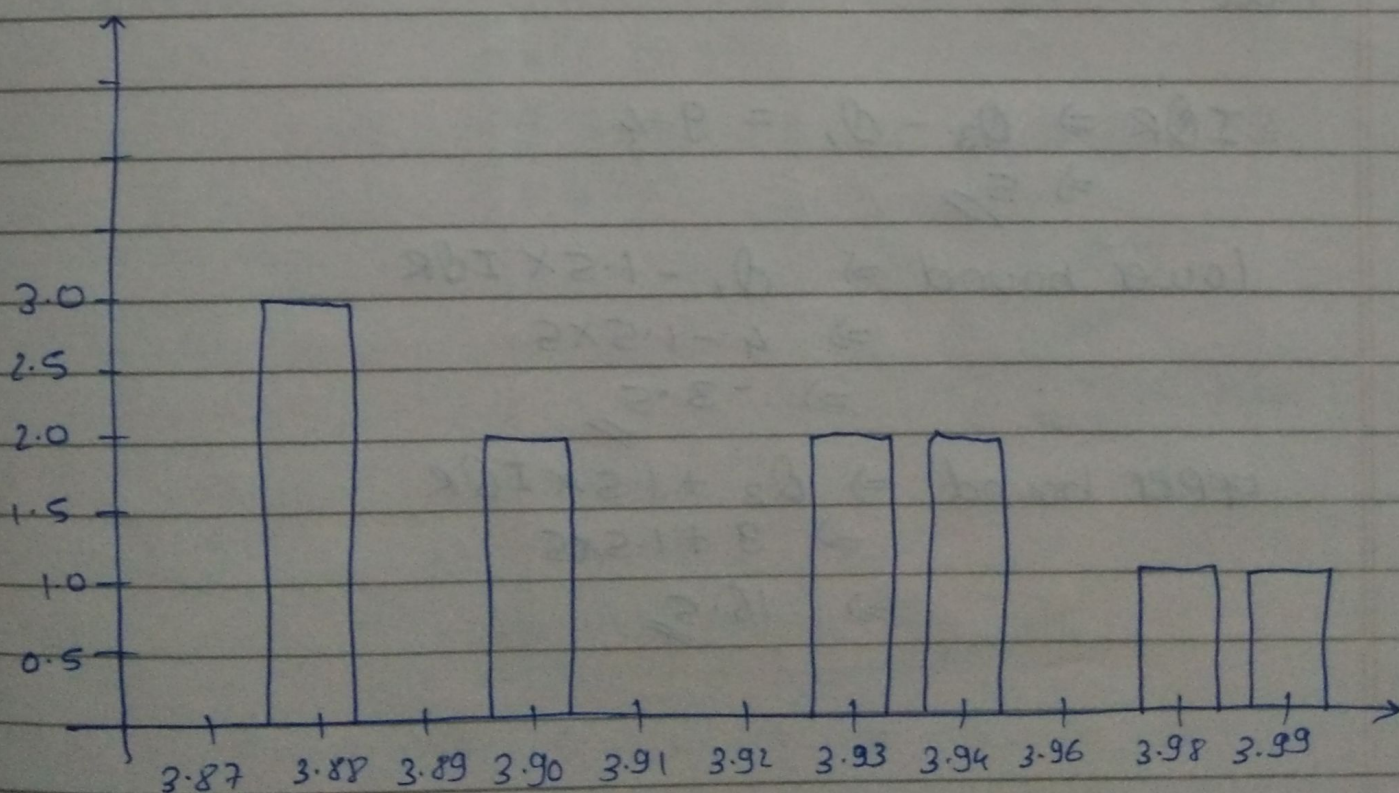
$$\therefore \text{skewness} \Rightarrow \frac{3(3.95 - 3.94)}{0.034} \approx 0.088. \text{ (true)}$$

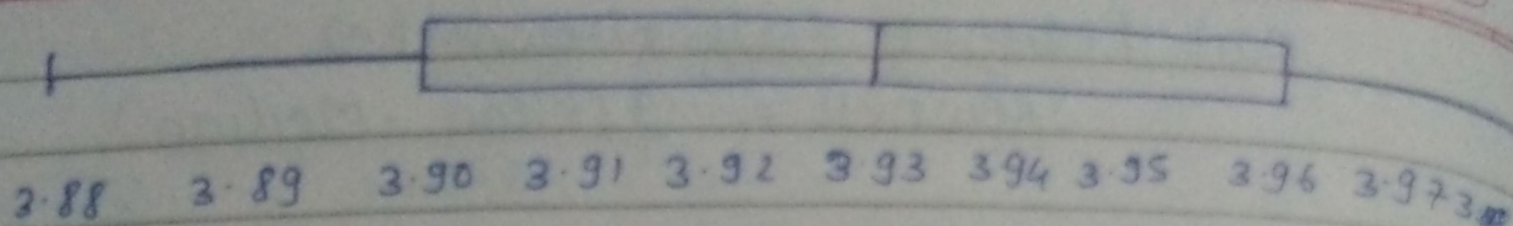
$$\text{Bowley's coeff. of skewness} \Rightarrow \frac{Q_3 + Q_1 - 2 \times \text{Median}}{Q_3 - Q_1}$$

$$\Rightarrow \frac{3.96 + 3.90 - 2 \times 3.94}{3.96 - 3.90} \Rightarrow \frac{0.08}{0.06} \approx 1.33. \text{ (truly skewed)}$$

$$\text{kurtosis} \Rightarrow \frac{\sum (x_i - \text{Mean})^4}{n(\text{S.D.})^4} - 3$$

$$\approx 0.5 \text{ (mesokurtic)}.$$





Q8]

Following data,

5, 7, 14, 0, 7, 9, 4, 10, 0, 8

Find range, histogram, box plot, mean, mode, Karl Pearson's coeff. of skewness, Bowley's coeff. of skewness, coeff. of Kurtosis. Hence state tively / vely skewed or is symmetric. Also state platykurtic / meso / lept.

→ Range $\Rightarrow$ Max - Min
 $\Rightarrow 14 - 0$
 $\Rightarrow 14$

Min $\Rightarrow 0$

$Q_1 \Rightarrow$ Median of 1st half (0, 0, 4, 5, 7) $\Rightarrow 4$

$Q_2 \Rightarrow$ Median $\Rightarrow 7$

$Q_3 \Rightarrow$ Median of 2nd half (7, 7, 8, 9, 10, 14) $\Rightarrow 9$

Max $\Rightarrow 14$

IQR $\Rightarrow Q_3 - Q_1 = 9 - 4$
 $\Rightarrow 5$

lower bound $\Rightarrow Q_1 - 1.5 \times \text{IQR}$
 $\Rightarrow 4 - 1.5 \times 5$
 $\Rightarrow -3.5$

upper bound $\Rightarrow Q_3 + 1.5 \times \text{IQR}$
 $\Rightarrow 9 + 1.5 \times 5$
 $\Rightarrow 16.5$

$$\text{Mean} \Rightarrow \frac{5+7+14+0+7+9+4+10+0+8}{10} \Rightarrow \frac{64}{10} \Rightarrow 6.4$$

$$\text{Mode} \Rightarrow 7$$

$$\text{skewness} \Rightarrow \frac{3(\text{Mean} - \text{Median})}{\text{S.D.}}$$

$$\text{var.} \Rightarrow \frac{\sum (x_i - \text{Mean})^2}{n}$$

$$\Rightarrow \frac{157.6}{10} \Rightarrow \underline{\underline{15.76}}$$

$$\text{S.D.} \Rightarrow \sqrt{15.76} \Rightarrow \underline{\underline{3.97}}$$

$$\therefore \text{skewness} \Rightarrow \frac{3(6.4 - 7)}{3.97} \approx \frac{-1.8}{3.97} \approx \underline{\underline{-0.45}}$$

(negatively skewed)

$$\therefore \text{Bowley's skewness} \Rightarrow \frac{d_3 + d_1 - 2 \times \text{Median}}{d_3 - d_1}$$

$$\Rightarrow \frac{9 + 4 - 2 \times 7}{9 - 4}$$

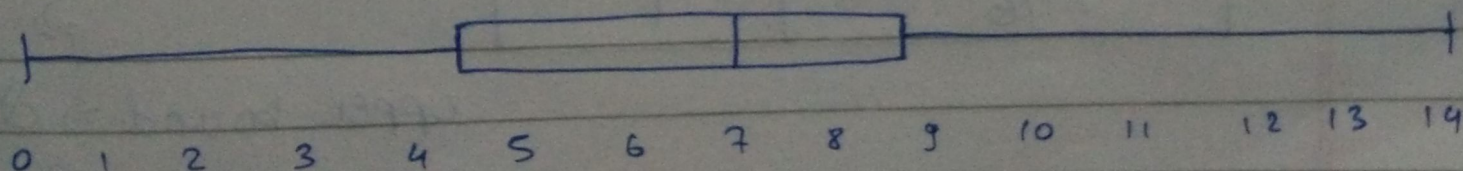
$$\Rightarrow \frac{-1}{5} \Rightarrow \underline{\underline{-0.2}}$$

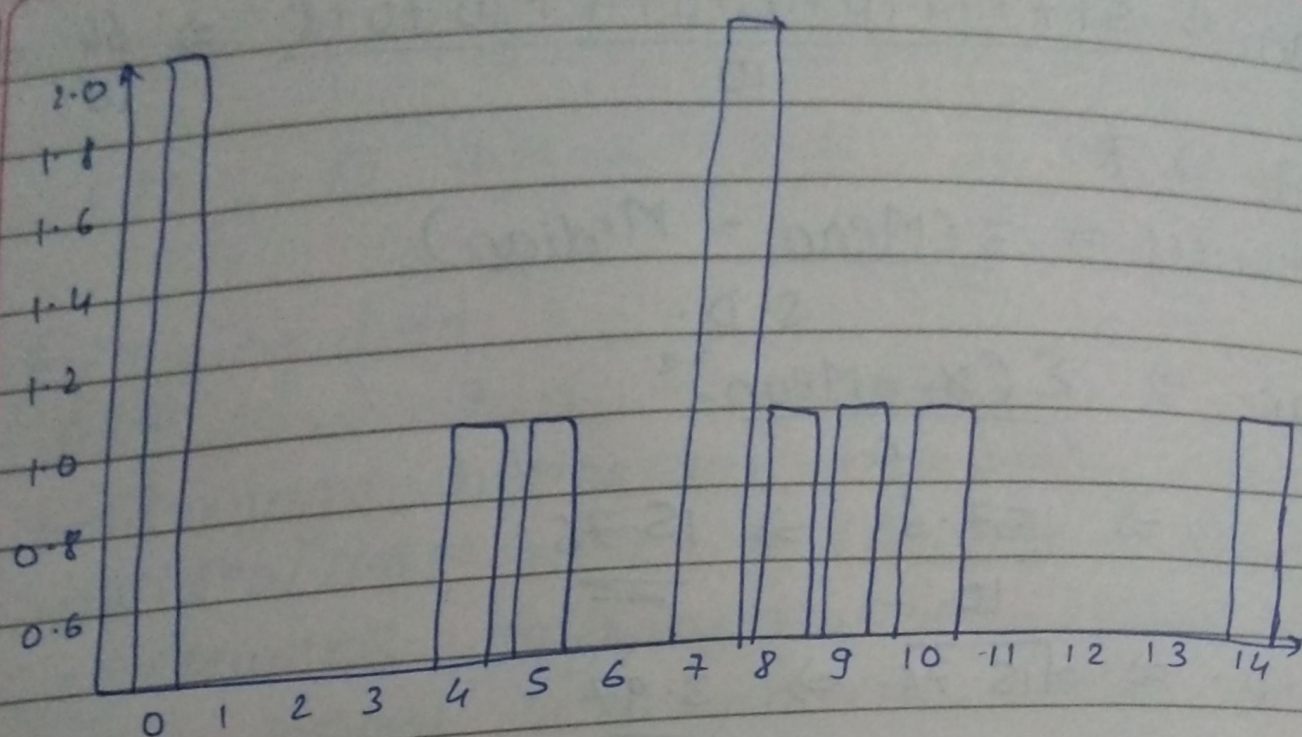
$$\text{Kurtosis} \Rightarrow \frac{\sum (x_i - \text{Mean})^4}{n \times (\text{S.D.})^4} - 3$$

$$\Rightarrow \frac{69.584}{(3.97)^4} - 3 \approx \underline{\underline{0.278 - 3}}$$

$$\approx \underline{\underline{-2.722}}$$

(Platy Kurtic)





Q12] Following data,
 8, 5, 12, 3, 9, 10, 6, 12, 8, 8, 4, 16, 10, 11, 7, 7, 3, 5, 9
 Create freq. table. Histogram box plot. Find mean
 mode, Karl Pearson's coeff. of skew, Bowley's
 kurtosis. Hence, +vely / -vely skewed or symmetric.
 Platykurtic / meso / leptokurtic.

→

No. of cars sold	Freq.
3	2
4	1
5	2
6	1
7	2
8	4
9	3
10	2
11	2
12	2
16	1

Min $\Rightarrow 3$

$Q_1 \Rightarrow 7$ (25^{th} %ile)

$Q_2 \Rightarrow$ Median $\Rightarrow 9.5$

(avg of 9 & 10)

$Q_3 \Rightarrow 11$ (75^{th} %ile)

Max $\Rightarrow 16$.

$$IQR = Q_3 - Q_1$$

$$= 11 - 7$$

$$\Rightarrow 4 //$$

$$\text{Lower bound} \Rightarrow Q_1 - 1.5 IQR$$

$$\Rightarrow 1 //$$

$$\text{Upper bound} \Rightarrow Q_3 + 1.5 IQR$$

$$\Rightarrow 17 //$$

Mean $\Rightarrow 8.15$

Mode $\Rightarrow 8$

S.D. $\Rightarrow 3.36$

$\therefore$ skewness $\Rightarrow \frac{3(8.15 - 8)}{3.36} = 0.1384$ (truly skewed)

$\therefore$ Bowley's $\Rightarrow \frac{Q_3 - Q_1}{Q_3 + Q_1}$

$\Rightarrow 0.33$

kurtosis $\Rightarrow 0.12$ (truly platykurtic)

